Skills Summary

Integer Exponent Laws: Product and Quotient

Use this reference while completing your worksheet or when studying.

The one idea behind every rule below: a power counts equal factors, so $2^3 \cdot 2^4$ is three factors of 2 set beside four more — seven in all. Every law here is bookkeeping on that count. Bases never change; only the exponent moves.

Skill 1 — Product of powers: add the exponents

Rule: $b^m \cdot b^n = b^{m+n}$ (same base only)
Why: m factors beside n factors is $m+n$ factors.

Example: Simplify $x^3 \cdot x^5$.
Step 1. The bases match and the powers are multiplied, so this is the product law — the factor counts add.
Step 2. $x^{3+5} = \boldsymbol{x^8}$

Try it: Simplify $m^5 \cdot m^4$. check: $\boldsymbol{m^9}$

Skill 2 — Quotient of powers: subtract the exponents

Rule: $\dfrac{b^m}{b^n} = b^{m-n}$ $(b \neq 0)$
Why: each factor below cancels one factor above, so the exponents subtract. Coefficients divide separately: $\dfrac{12x^5}{4x^2} = 3x^3$.

Example: Simplify $\dfrac{x^7}{x^2}$.
Step 1. Same base in a quotient, so cancelling pairs is what the subtraction records — top exponent minus bottom exponent.
Step 2. $x^{7-2} = \boldsymbol{x^5}$

Try it: Simplify $\dfrac{10^8}{10^5}$ as a power of 10. check: $\boldsymbol{10^3}$

Skill 3 — Zero and negative exponents

Rules: $b^0 = 1$ and $b^{-n} = \dfrac{1}{b^n}$ $(b \neq 0)$

Neither is arbitrary: subtracting equal exponents gives b^0 for an expression that obviously equals 1, and subtracting a larger exponent from a smaller one leaves the extra factors in the denominator.

Example: Simplify $\dfrac{5^3}{5^5}$.

Step 1. Same base, so subtract — and keep the sign of the result instead of flipping the subtraction to avoid it.

Step 2. $5^{3-5} = \mathbf{5^{-2}}$, which is $\dfrac{1}{5^2} = \dfrac{1}{25}$

Try it: Simplify $\dfrac{2^3}{2^6}$ as a power of 2 with a negative exponent. check: $\mathbf{2^{-3}}$

(!) Watch out: Adding exponents is not multiplying them. $2^5 \cdot 2^3 = 2^8 = 256$, never 2^{15}. And a negative exponent does not make a negative number: 4^{-3} is a small positive fraction.